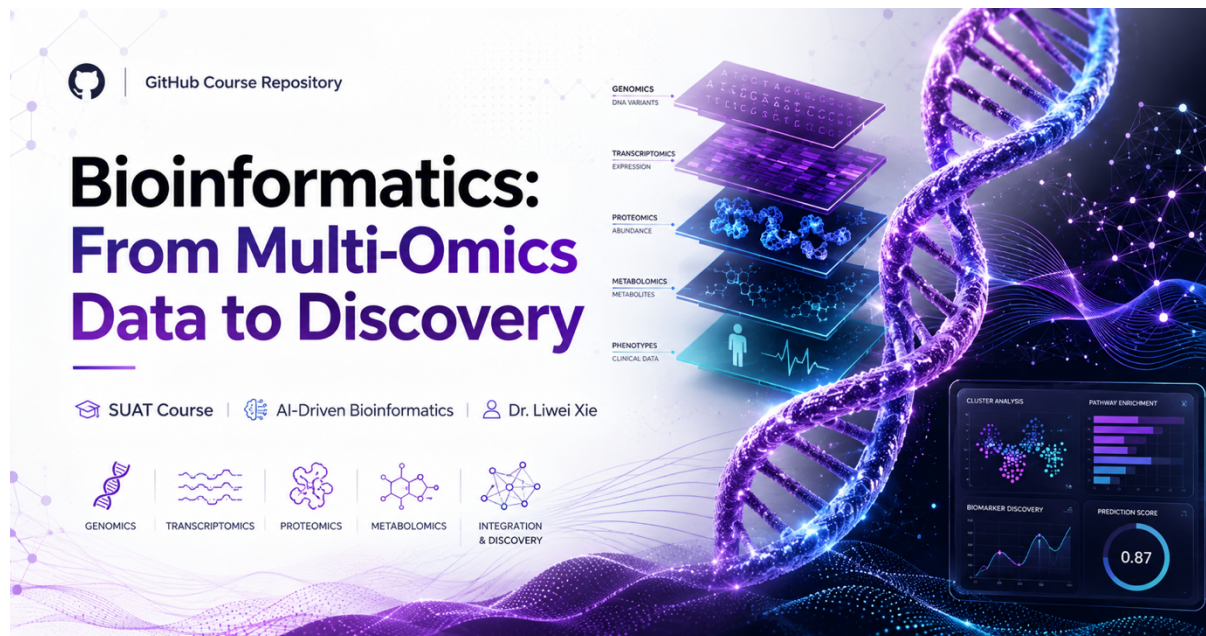




Course Syllabus — Fall Semester 2026

Bioinformatics: From Multi-Omics Data to Discovery



I. Course Information

Field	Details
Course Title (Chinese)	生物信息：从多组学数据到科学发现
Course Title (English)	Bioinformatics: From Multi-Omics Data to Discovery
Semester/Time	Year 3, Fall Semester (2026); Wen 10:15-11:40
Credits / Total Hours	2 Credits / 32 Hours
Course Category	Core Major Course
Target Students	3rd-year undergraduates in Synthetic Biology / Life Sciences
Offering Department	Faculty of Synthetic Biology
Language of Instruction	50% Chinese / 50% English (lectures); 100% English (PPT, reading materials)
Prerequisites	Calculus; Introduction to Life Sciences; Biochemistry
Assessment Type	Assessment and project-based
Grading System	Five-tier (A+/A/A-/B+/B/B-/C+/C/D+/D/F; D and above = pass, F=failed)
Course Coordinator	Instructor: Prof. Liwei Xie, Ph.D. Email: xieliwei@suat-sz.edu.cn Office Hours: Wen 8:30 AM – 9:30 AM, or by appointment. Address: TianShu II (BLD 19B) Office 318
Instructors	Prof. Liwei Xie (谢黎炜); Prof. Guangchuang Yu (余光创, Southern Medical University)

II. Course Description

This course is designed for third-year undergraduates in Synthetic Biology, Life Sciences, and related programs. It is an innovative bioinformatics introductory but applicable course that integrates cutting-edge Artificial Intelligence (AI) technology with multi-omics data analysis. Unlike traditional bioinformatics courses that emphasize the memorization of coding syntax, this course focuses on training students to utilize Large Language Models (LLMs) as intelligent assistants for code generation, debugging, imaging and data analysis and interpretation.

The curriculum covers core areas including Genomics, Transcriptomics, Proteomics, Metabolomics, Microbiome, Single Cell and Spatial transcriptomics studies. Adopting a **"Background Theory + LLM Practice"** teaching model, students will learn to handle real-world biological data by using a R-based multiomics analysis platefor. The central aim is to simulate the complete scientific research workflow: from formulating scientific questions, data fetching and cleaning via LLM, statistical analysis and visualization, to final data interpretation

and manuscript writing. Through this course, students will master essential skills for biological big data mining in the AI era.

III. Textbooks and References

The course draws on a combination of custom-built instructional materials and internationally recognized references:

#	Resource	Details
1	Custom Lecture Notes / PPT	Tailored materials on LLM Prompt Engineering and multi-omics analysis workflows
2	Primary Reference Textbook	<i>Bioinformatics and Functional Genomics</i> , 3rd ed. (Jonathan Pevsner)
3	Programming Reference	<i>R for Data Science</i> (Hadley Wickham)
4	Online Resources	NCBI and GEO database official documentation; official best-practice guides for ChatGPT, Claude, DeepSeek, Agent Hub, EasyMultiProfiler

IV. Learning Objectives

The course is structured around three tiers of learning outcomes:

Objective 1 — Knowledge: Students will develop a deep understanding of multi-omics data structures (genomics, transcriptomics, proteomics/metabolomics, and microbiome), their biological significance, and the logic underlying each analytical pipeline.

Objective 2 — Competency: Students will master Prompt Engineering, enabling them to fluently guide LLMs to generate high-quality R code for specific biological problems. They will also acquire the ability to retrieve and clean data from major public repositories.

Objective 3 — Critical Thinking & Integrity: Students will cultivate critical thinking skills, learning to identify and correct AI-generated errors ("hallucinations"), and will be capable of independently completing the full cycle from data analysis to scientific manuscript writing.

V. Course Ideology and Ethics

This course integrates values education into technical training through three dimensions:

Scientific Integrity: When using AI to assist in research, the course places strong emphasis on data authenticity and research integrity. Students are educated to be vigilant about AI "hallucinations" and not to blindly follow algorithmic results, thereby cultivating a rigorous and truth-seeking scientific attitude.

Innovation Mindset: By introducing the latest LLM technologies, the course breaks the traditional rote-learning model and encourages students to embrace new tools, fostering innovative thinking for solving complex biological problems in the AI era.

National Pride and Academic Confidence: In the Genomics and Microbiome chapters, the course highlights the outstanding contributions of Chinese scientists to the Human Genome

Project, rice genome sequencing, and human microbiome research (e.g., achievements by BGI and affiliated institutions), strengthening students' national pride and academic self-confidence.

VI. Weekly Schedule and Learning Objectives

Week	Topic	Hours	Teaching Method	Learning Objectives
1 (26 Aug)	Introduction & AI Fundamentals (Prompting for Bio)	2	Lecture + Demo	2, 3
2 (2 Sep)	Scientific Question Formulation & Literature Mining with AI	2	Lecture + Discussion + Practice	1, 3
3 (9 Sep)	Data Fetching & Cleaning: Public Repositories (GEO/NCBI)	2	Lecture + Practice + Exercises	2
4 (16 Sep)	Genomics: Sequencing Principles & Variant Theory and Analyzing Data with LLM	2	Lecture	1
5 (23 Sep)	Transcriptomics: RNA-Seq Principles & DE Analysis with DESeq2	2	Lecture + Lab/Practice	2
6 (30 Sep)	Microbiome: 16S & Metagenomics Overview and Diversity Analysis Practice	2	Lecture	1
7 (14 Oct)	Functional Omics: Proteomics & Metabolomics Strategies	2	Lecture + Lab/Practice	2, 3
8 (21 Oct)	Single Cell and Spatial Transcriptomics (Guest: Prof. Guangchuang Yu, Southern Medical University)	2	Lecture	1
9 (28 Oct)	Multi-Omics Integration & Pathway Enrichment (KEGG/GO)	2	Lecture + Lab/Practice	2
10 (4 Nov)	Data Visualization & Polishing (Advanced ggplot2 via AI)	2	Lecture + Discussion	1
11 (11 Nov)	Research Examples of Multi-Omics Data Analysis	2	Lecture + Lab/Practice	1, 2
12 (18 Nov)	Interpretation & Writing I: From Plots to Text	2	Lecture + Practice	2
13 (25 Nov)	Interpretation & Writing II: AI Polishing & Critiquing	2	Lecture + Discussion + Writing	3

Week	Topic	Hours	Teaching Method	Learning Objectives
14 (2 Dec)	Final Project Presentation & Defense I	2	Discussion / Presentation	3
15 (9 Dec)	Final Project Presentation & Defense II	2	Discussion / Presentation	1, 2, 3
16 (16 Dec)	Course Summary & Future Perspectives 课	2	Lecture	1, 3

Total Lecture Hours: 32 (each session = 2 × 40-minute classes)

VII. Lab and Practice Projects

The course includes four structured practical projects totaling 12 hours, each designed to simulate a real-world research scenario:

Project	Title	Hours	Content	Objectives
1	Build Your AI Bioinfo Assistant	1	Configure RStudio environment; test the effect of different prompts on code generation quality; complete a warm-up task of "plotting a differential analysis heatmap"	1, 2, 3
2	Differential Expression Analysis of Transcriptomic Data	2	Use LLM to write scripts; download TCGA or GEO data; perform normalization; filter differentially expressed genes; plot volcano plots and heatmaps	1, 2, 3
3	Gut Microbiome Diversity Exploration	2	Input OTU table; use LLM to generate Phyloseq analysis pipeline; calculate and visualize Alpha/Beta diversity	1, 2, 3
4	Multi-Omics Integration & Manuscript Writing	7	Select a specific scientific question (e.g., "metabolic mechanism of a drug"); integrate at least two omics datasets (e.g., transcriptomics + metabolomics); complete full analysis pipeline; use AI to assist in writing a mini-paper with Abstract, Methods, Results, and Discussion sections	1, 2, 3

VIII. Assessment and Grading

The course employs a process-oriented assessment model that emphasizes practical application and critical thinking over rote memorization.

Component	Assessment Criteria	Weight
Homework	Quality of project submissions and homework	10%
Class Participation/Quiz	Engagement in discussions, in-class exercises, Q&A and weekly quiz	20%
Final Project Presentation & Defense	Clarity, depth, and scientific rigor of the oral presentation, defense and Q&A	30%
Final Project Report	Completeness, analytical depth, and writing quality of the written report	40%
Total		100%

Grading Scale:

Grade	Range	Status
A+ / A / A-	Excellent	Pass
B+ / B / B-	Good	Pass
C+ / C	Satisfactory	Pass
D+ / D	Passing	Pass
F	Failing	Fail

IX. Course Policies

Attendance: Regular attendance is expected. Students are responsible for all material covered in class. Absences exceeding 20% of total class hours may result in course failure.

Academic Integrity: All submitted work must be the student's own. While AI tools (LLMs) are explicitly encouraged as learning aids in this course, students must disclose their use and demonstrate genuine understanding of the AI-generated content. Submitting AI-generated work without critical engagement or attribution constitutes academic dishonesty.

AI Tool Usage: Students are encouraged to use ChatGPT, Claude, DeepSeek, and similar LLMs as coding and writing assistants. However, students must be able to explain, verify, and critically evaluate all AI-generated outputs. Uncritical reliance on AI without understanding is strongly discouraged.



Late Submission Policy: Project reports submitted after the deadline will be penalized 10% per day, up to a maximum of 50%. Reports more than 5 days late will not be accepted without prior approval. Weekly homework submitted after the deadline will receive ZERO score.

Communication: Students are encouraged to communicate with the instructor during class and office hour. Students in each group must schedule at least one time office meeting to discuss the group project selection.

X. Instructor Biography

Dr. Liwei Xie (谢黎炜) is a tenured Associate Research Professor and Distinguished Professor at the Faculty of Synthetic Biology, Shenzhen University of Technology. He also holds appointments as a Researcher at the Guangdong Institute of Microbiology, Guangdong Academy of Sciences, and as a Distinguished Professor in the Department of Endocrinology and Metabolism at Zhujiang Hospital, Southern Medical University.

Dr. Xie received his B. Eng. in Automation from Xi'an University of Science and Technology (2007) and his B.S. in Biological Sciences from the State University of New York at Buffalo (2009). He earned his Ph.D. in Nutritional Sciences from the University of Florida (2013), followed by postdoctoral training at the University of Michigan Medical School and the Center for Molecular Medicine at the University of Georgia (2013–2017).

His research focuses on gut microbiota-mediated inter-tissue crosstalk and human health, encompassing systemic trace element metabolism, physiological regulation of muscle and adipose stem cells, and the mechanistic role of gut microbiota–host nutritional metabolic interactions in metabolic diseases. In the past five years, he has published over 50 high-impact research papers with more than 3,000 citations (single-paper maximum exceeding 200 citations) in journals including *The Innovation* (IF: 32.2), *Journal of Clinical Investigation* (IF: 17.884), *Advanced Science* (IF: 15.608), *Journal of Cachexia, Sarcopenia and Muscle* (IF: 12.29), and *SCIENCE CHINA-Life Sciences* (IF: 9.1). He holds 6 Chinese invention patents and 4 international patents, and has been awarded multiple grants from the National Natural Science Foundation of China and the Guangdong Outstanding Youth Fund.

XI. Important Dates (Fall 2026)

Event	Date
First Day of Class	Week 1, 26 Aug 2026
Final Presentation (Week 14–15)	Weeks 14–15
Last Day of Class	Week 16